PATIN / TERO!

SECTION 3 — CORE GAMEPLAY MECHANICS

Game Design Document | Reconstructed Draft v1.0 | Review copy

Section 3 defines the rules governing the two arenas, unit movement, programmed commands, tagging, and battle resolution. It builds on Section 2 and incorporates the latest decisions about fixed deployment points and persistent 3D arenas. This is a reconstructed section rather than a verbatim copy of the earlier GDD. Mechanics requiring confirmation are identified as provisional.

3.1 Gameplay Overview

PATIN / TERO! is a deterministic, grid-based strategy auto-battler in which players program their characters before a match and observe their strategies execute automatically. Each player simultaneously controls an offensive formation of runners and a defensive formation of guards. Both formations operate on separate 4×8 arenas.

The objective is to create effective offensive movement patterns while preventing the opponent's runners from completing their own objectives. Planning Phase: players purchase characters, deploy units to valid starting points, and select and order their moveset cards. Battle Phase: both players' programmed formations execute automatically, without intervention. Identical initial states and commands must produce identical outcomes.

3.2 Dual-Arena System

Arena	Friendly units	Opposing units	Objective
Offense	Up to 3 runners	4 guards	Complete a round trip
Defense	4 guards	Up to 3 runners	Prevent a round trip

Both arenas are displayed side by side using a slightly isometric 3D perspective. The arenas used during planning are the exact same physical boards used during battle. Only the units, overlays, available interactions, and simulation state change.

3.2.1 Arena Dimensions

Each arena consists of four columns and eight rows, for a total of 32 grid cells. The logical grid uses zero-based coordinates: columns 0–3 and rows 0–7. Row 0 is the runner starting side; row 7 is the opposite side and turnaround point. Authoritative grid coordinates are independent of the 3D presentation.

3.3 Runner Mechanics

Runners are offensive units tasked with crossing the grid and returning to their starting side. Each offensive formation can deploy a maximum of three runners.

3.3.1 Runner Deployment

Runners can only be deployed on four designated starting points along row 0. Each starting point can contain only one runner. Runners cannot be deployed on the defense arena or on other rows.

3.3.2 Runner Objective

A runner begins on row 0, advances toward row 7, reaches row 7 without being tagged, reverses its forward-facing direction, and returns to row 0 without being tagged. This completes a round trip.

3.3.3 Runner Direction

Runners begin facing toward row 7. Upon reaching row 7, their forward direction reverses so later forward commands move toward row 0. Left and right commands are interpreted relative to the runner's current facing direction.

3.3.4 Runner Elimination

A runner is tagged when it enters a valid guard interception condition. A tagged runner is removed from the active arena and cannot complete its round trip. For the MVP, runners have no health points, damage reduction, or respawning.

3.4 Guard Mechanics

Guards are defensive units tasked with preventing opposing runners from completing round trips. Each defensive formation contains up to four guards, divided into horizontal and center roles.

3.4.1 Horizontal Guards

Guard	Assigned row	Patrol direction
Horizontal Guard 1	2	Left / right
Horizontal Guard 2	4	Left / right
Horizontal Guard 3	6	Left / right

Horizontal guards cannot leave their assigned patrol rows. Their starting positions are selected from predefined deployment anchors along those rows. During battle, their movement is controlled entirely by equipped moveset cards.

3.4.2 Center Guard

The center guard patrols the vertical line between columns 1 and 2, moving up and down. It may be deployed only on predefined starting anchors belonging to that line. Its physical position is represented on a grid line rather than at a standard cell center.

3.4.3 Guard Restrictions

Guards cannot leave their patrol lines, enter the friendly offense arena, change equipped cards during battle, or react to opponents unless the reaction is explicitly encoded by a command. Guards do not use reactive AI.

3.5 Movement Command System

Every unit moves according to predefined commands contained in signature moveset cards. Commands execute sequentially and automatically.

3.5.1 Runner Commands

Command	Name	Behavior
F	Forward	Move one row in facing direction
L	Left	Move one column left relative to facing
R	Right	Move one column right relative to facing
DL	Diagonal Left	Forward one row and left one column
DR	Diagonal Right	Forward one row and right one column
S	Stay	Remain stationary for 500 ms
T	Taunt	Perform TAUNT for 750 ms

3.5.2 Guard Commands

Command	Name	Behavior
L	Left	Move one column left
R	Right	Move one column right
U	Up	Move one row upward along center line
D	Down	Move one row downward along center line
S	Stay	Remain stationary for 500 ms
C	Scare	Perform SCARE for 750 ms

Horizontal guards use horizontal commands; center guards use vertical commands. Each command has a defined execution duration. The simulation uses a shared deterministic clock rather than visual animation completion. Commands blocked by boundaries or patrol restrictions follow a consistent fallback rule.

3.6 Equipment-Based Signature Moveset System

Each character possesses exactly three unique signature moveset cards. Every signature card contains four commands. Players cannot edit commands within cards; they equip cards and arrange execution order.

Star level	Card capacity	Rule
1 star	1	Equip one of three signature cards
2 stars	2	Equip two distinct signature cards
3 stars	3	Equip all three signature cards

Cards are character-specific and cannot be transferred between different characters. When multiple cards are equipped, they execute in selected order and loop after the final command. For example, Card A: F → DL → F → DR, followed by Card B: F → F → L → F, then back to Card A. Execution repeats until elimination, objective completion, or arena resolution.

Strategic effectiveness depends on character selection, signature-card combinations, ordering, starting positions, command timing, and patrol intersections.

3.7 Deterministic Simulation

The authoritative simulation must be independent of rendering frame rate, camera movement, visual effects, and input latency during battle. Units begin from validated positions, execute fixed ordered commands using a shared clock, and resolve movement and collisions using consistent rules. Visual animations interpolate authoritative state but never change outcomes.

3.7.2 Simultaneous Movement

Multiple units may execute overlapping commands. The simulation must consistently resolve movement and tagging, including cases where a runner and guard cross an intersection at nearly the same moment. Precise simultaneous-collision priority rules must be defined and tested before implementation.

3.8 Tagging and Collision System

Tagging is the primary defensive mechanic. Guards eliminate runners by intercepting their movement across designated patrol lines. The simulation evaluates crossings automatically; no real-time player input is required.

3.8.1 Horizontal Guard Tagging

A runner may be tagged when it crosses a horizontal guard's patrol line at a position and time covered by that guard. The simulation must evaluate movement paths, not just final cell positions, including diagonal crossings.

3.8.2 Center Guard Tagging

A runner crossing the center line between columns 1 and 2 may be intercepted if the center guard occupies the relevant position at crossing time. Collision detection must account for crossings between cells.

3.8.3 Tagging Outcomes

On a successful tag, record the event, mark the runner eliminated, stop its program, remove it from active gameplay, and play a visual tagging effect. Remaining runners continue. A runner cannot be revived or redeployed during that battle.

3.8.4 Collision Resolution

Situation	Expected behavior
Cross occupied guard line	Evaluate tagging
Cross unoccupied guard line	Continue movement
Runner and guard move simultaneously	Use deterministic timing and priority
Two runners share a cell	Both may remain active unless another rule applies
Multiple guards intercept one runner	Record one elimination
Runner reaches row 7	Reverse forward direction
Runner returns to row 0	Register successful round trip

Exact interception timing, crossing priority, and simultaneous arrivals remain to be finalized.

3.9 Special Commands — TAUNT and SCARE

3.9.1 TAUNT

TAUNT (T) is runner-specific, lasts 750 ms, and keeps the runner stationary during execution. It introduces a deliberate timing window and may interact with guard SCARE behavior.

3.9.2 SCARE

SCARE (C) is guard-specific, lasts 750 ms, and keeps the guard stationary during execution. It creates a temporary defensive interaction potentially affecting interception.

3.9.3 Interaction

Both commands are timed, automatic, and resolved deterministically. Their effects and activation windows must be communicated through visual feedback. The precise counter relationship, range, and priority remain to be confirmed; no immunity, stun, or cancellation effect is assumed.

3.10 Movement Boundaries and Invalid Commands

Runners remain within columns 0–3 and rows 0–7; reaching row 7 reverses forward direction, and returning to row 0 completes the objective. Horizontal guards remain on their assigned patrol rows within columns 0–3. Center guards remain on their center patrol line and permitted vertical range.

Provisional MVP fallback: an invalid movement command consumes its normal duration while the unit stays at its current position; execution then proceeds to the next command. Confirm before implementation.

3.11 Battle Timing and Execution

Command type	Duration
STAY	500 ms
TAUNT	750 ms
SCARE	750 ms
Movement commands	Fixed duration — to be finalized

When both players lock formations, the game validates them, creates the initial simulation state, places units at anchors, initializes command indices, completes a countdown, and begins both arenas at the shared match start time. Each arena resolves independently; the match ends after both outcomes are available.

3.12 Arena Victory and Resolution Conditions

An offensive victory occurs when at least one runner reaches row 7 and returns to row 0 without being tagged. A successful trip is recorded with completion time. A defensive victory occurs when guards prevent a successful trip under the applicable resolution rule, such as eliminating all runners. Timeout treatment follows finalized tournament rules.

State	Description
Offense Victory	Runner completes a round trip
Defense Victory	Guards prevent successful round trip
Timeout	Arena reaches its time limit

State	Description
Resolved	Final arena outcome recorded

The exact arena time limit and timeout winner assignment remain to be confirmed.

3.13 Formation Locking

A locked formation contains deployed unit identities, star levels, assigned anchors, equipped signature cards, and card execution order. Validation ensures role-compatible anchors, no duplicate occupants, runner and guard limits, legal card counts, no duplicate equipped cards, and cards belonging to the selected character.

After lock, players cannot move units, change anchors, edit movesets, or purchase/merge for the current battle. The locked configuration becomes the authoritative initial battle state.

3.14 Persistent Planning and Battle Arenas

The same physical 3D arenas are used during planning and battle. Players drag units from the bench onto role-specific anchors. Runner anchors are on offense row 0; horizontal guard anchors lie on assigned defense rows; center guard anchors lie on the center vertical patrol line. Invalid drops return units to their previous valid positions.

Feature	Planning	Battle
Arena geometry	Visible	Same
Camera	Slightly isometric	Same orientation
Unit deployment	Enabled	Disabled
Shop	Modal available	Unavailable
Moveset configuration	Enabled	Disabled
Simulation	Stopped	Running
Movement preview	Available	Disabled
Battle HUD	Minimal	Active

3.15 Gameplay Feedback and Readability

Units visibly communicate direction, active command, and state. Animations are synchronized with simulation events. Tags produce recognizable effects and distinguish eliminated runners. TAUNT and SCARE have clear activation and expiration cues. Round trips and arena victories receive distinct scoring feedback.

3.16 Core Mechanics Design Goals

Deterministic Fairness: outcomes follow initial formations, commands, and fixed rules. Strategic Depth Through Simplicity: simple commands yield complex patterns. Meaningful Character Selection: distinct signature movesets shape composition. Readable Automated Battles: players understand movement and tagging without intervention. Consistent Planning and Execution: the same arenas and logical rules apply throughout.

Section 3 — Review Notes

Before final implementation, confirm exact movement-command durations, TAUNT/SCARE effects, simultaneous crossing/tagging priority, boundary fallback behavior, and arena timeout/victory rules. Section 9's deterministic simulation architecture depends on these definitions.